

Choline intake and risk of lethal prostate cancer: incidence and survival

Background: Meat, milk, and eggs have been inconsistently associated with the risk of advanced prostate cancer. These foods are sources of choline—a nutrient that may affect prostate cancer progression through cell membrane function and one-carbon metabolism. No study has examined dietary choline and the risk of lethal prostate cancer. **Objective:** Our objective was to examine whether dietary choline, choline-containing compounds, and betaine (a choline metabolite) increase the risk of lethal prostate cancer. **Design:** We prospectively examined the intake of these nutrients and the risk of lethal prostate cancer among 47,896 men in the Health Professionals Follow-Up Study. In a case-only survival analysis, we examined the postdiagnostic intake of these nutrients and the risk of lethal prostate cancer among 4282 men with an initial diagnosis of nonmetastatic disease during follow-up. Diet was assessed with a validated questionnaire 6 times during 22 y of follow-up. **Results:** In the incidence analysis, we observed 695 lethal prostate cancers during 879,627 person-years. Men in the highest quintile of choline intake had a 70% increased risk of lethal prostate cancer (HR: 1.70; 95% CI: 1.18, 2.45; P-trend = 0.005). In the case-only survival analysis, we observed 271 lethal cases during 33,679 person-years. Postdiagnostic choline intake was not statistically significantly associated with the risk of lethal prostate cancer (HR for quintile 5 compared with quintile 1: 1.69; 95% CI: 0.93, 3.09; P-trend = 0.20). **Conclusion:** Of the 47,896 men in our study population, choline intake was associated with an increased risk of lethal prostate cancer.